

Spectraplakin cooperates with noncentrosomal microtubule regulators to orient dendritic microtubules in *Drosophila*

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eLife Assessment

Davies et al. present a **valuable** study proposing that Shot can act as a molecular linker between microtubules and actin during dendrite pruning, suggesting an intriguing role in non-centrosomal microtubule organization. However, the experimental evidence is **incomplete** and does not robustly support these claims, and the lack of a cohesive model connecting the findings weakens the overall impact. While the data suggest that Shot, actin, and microtubule nucleation contribute to dendritic pruning, their precise interplay remains unresolved.

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Abstract

The differential microtubule organization in axons and dendrites underlies neurite-specific transport and morphogenetic processes like developmental neurite pruning, but how it arises is incompletely understood. While axonal "plus end-out" microtubules just grow from the soma, the origin of the dendritic "plus end-in" microtubules is more difficult to explain. A transient microtubule organizing center (MTOC) was recently detected at dendrite tips in *C. elegans*, but the players and generalizability are unclear. In a survey for factors involved in dendrite pruning of *Drosophila* sensory neurons, we found the spectraplakin Short stop (Shot), an actin-microtubule crosslinker known to bundle axonal microtubules. In dendrites, however, Shot is required for proper "plus end-in" microtubule organization. During early dendrite development, Shot localizes to dendrite tips in an actin-dependent manner, providing a potential mechanism for dendrite tip recognition. Our data show that Shot cooperates with a functional MTOC containing the small GTPase Rab11. Localized MTOC activity may commonly underlie dendritic microtubule organization.

Introduction

Microtubules are long polar tube-like polymers composed of α/β -tubulin dimers that provide mechanical stability to cells and act as tracks for motor-based transport. Tubulin dimers are preferentially added, and also lost, at microtubule plus ends, whereas microtubule minus ends are often anchored at microtubule organizing centers (MTOCs), structures of diverse composition that nucleate and anchor microtubules. In dividing cells, centrosomes are the main MTOCs, but many non-dividing or differentiated cells have less well characterized non-centrosomal (nc)MTOCs (Akhmanova & Kapitein, 2022 [↗](#)).

Given their intrinsic polarity, microtubules predefine neuronal polarity. In axons, all microtubules are oriented with their plus ends towards the distal synapses, thus enabling axon-specific transport via kinesin-1. In contrast, a large fraction of dendritic microtubules are oriented with their plus ends towards the soma ("plus end-in"), enabling dendrite-specific transport via other motors including dynein. This orientation is particularly evident in invertebrates, where dendritic microtubules in primary dendrite branches almost uniformly adopt the plus end-in orientation. Despite the recognized importance of microtubule organization for neuronal function, it is still poorly understood how neuronal microtubule organization is achieved and maintained.

The differential organization of neuronal microtubules also underlies important developmental processes. The peripheral nociceptive class IV dendritic arborization (c4da) neurons of *Drosophila* larvae prune their long and branched sensory dendrites at the onset of metamorphosis. Pruning involves severing of proximal dendrite regions and subsequent degeneration of distal fragments (Rumpf et al, 2017 [↗](#); Furusawa & Emoto, 2021 [↗](#)). The presumptive severing sites are marked by the local loss of microtubules, which predisposes them for mechanical tearing (Krämer et al, 2023 [↗](#)). This local microtubule disassembly is initiated by an increase in microtubule dynamics at the onset of the pupal stage (Herzmann et al, 2017 [↗](#)). Due to their uniform plus end-in orientation, dendritic microtubules can then shrink from their plus ends in a coordinated fashion, leading to their local loss in proximal dendrite regions (Herzmann et al, 2018 [↗](#); Rumpf et al, 2019 [↗](#)). In support of this model, conditions that cause misorientation of dendritic microtubules bring about pruning defects. For example, pruning defects are associated with mutations in the kinesin-2 motor, which guides newly polymerized microtubules along preexisting ones, thus ensuring the identical orientation (Mattie et al, 2010 [↗](#); Herzmann et al, 2018 [↗](#)). Microtubule orientation and pruning defects are also caused by loss of the minus end binding protein patronin (Wang et al, 2019 [↗](#)), the microtubule polymerase/nucleation factor Minispindles (MSPs)/XMAP-215 (Tang et al, 2020 [↗](#)) and the serine/threonine phosphatase PP2A (Rui et al, 2020 [↗](#); Wolterhoff et al, 2020 [↗](#)), but the underlying mechanisms are not always clear. An additional mechanism for dendritic microtubule orientation was described in *C. elegans* PVD neurons, where the localization of dendritic MTOCs is important for the establishment of uniform plus end-in organization. Here, a prominent ncMTOC resides at the tips of growing dendrites, such that microtubules can only grow towards the soma (Liang et al, 2020 [↗](#); Harterink et al, 2018 [↗](#)). This ncMTOC was shown to reside on Rab11-positive recycling endosomes, and Rab11 was important for microtubule orientation (Liang et al, 2020 [↗](#)).

Spectraplakins are a family of actin/microtubule cytoskeletal crosslinker proteins with important roles in microtubule organization in differentiated cells (Sanchez et al, 2021; Sun et al, 2019 [↗](#); Voelzmann et al, 2017 [↗](#)). Anchored to the actin cortex via an N-terminal actin binding domain, they can recruit microtubules either directly via a GAS2-related domain, or via binding sites for end binding proteins (Noordstra et al, 2016 [↗](#); Ning et al, 2016 [↗](#); Nashchekin et al, 2016 [↗](#)) (Slep et al, 2005 [↗](#)). The *Drosophila* spectraplakins Short stop (Shot) is known to bundle and stabilize microtubules in axons (Alves-Silva et al, 2012 [↗](#); Okenve-Ramos et al, 2024 [↗](#)).

Here, we describe an important role for Shot in dendrites. Loss of Shot causes pruning defects caused by mixed orientation of dendritic microtubules. During early dendrite growth stages, Shot localizes to dendrite tips in an actin-dependent manner where it recruits factors cooperating with an early-acting, Rab11-dependent MTOC. Our data provide a mechanism for MTOC tip recruitment for the establishment of oriented microtubule arrays in dendrites.

Results

Shot is required for c4da neuron dendrite pruning

C4da neurons have long and branched dendrites at the third instar larval (L3) stage (**Fig. 1** [A](#)) that are pruned at the onset of the pupal stage, such that by 16 hours after puparium formation (h APF), they have lost all their dendrites (**Fig. 1** [A'](#), G - I). In order to identify new regulators of c4da neuron dendrite pruning, we expressed candidate RNAs in c4da neurons under *pickpocket-GAL4* and assessed their effects on dendrite morphology and pruning. This approach identified a dsRNA line targeting *Shot*, which encodes the sole Drosophila member of the spectraplakins, a family of large adaptor proteins with binding sites for actin and microtubules. C4da neurons expressing *shot* dsRNA did not display major dendritic phenotypes at the larval stage (**Fig. 1** [B](#)), even though their dendritic field coverage seemed somewhat reduced compared to control neurons. In contrast, dendrites could still be seen attached to the soma at 16 h APF in almost half the c4da neurons expressing *shot* dsRNA (**Fig. 1** [B'](#), G - I), indicative of pruning defects. In order to confirm this result, we used Mosaic Analysis with a Repressible Cell Marker (MARCM) to generate c4da neuron clones homozygous for the loss-of-function allele *shot*³ (Kolodziej et al, 1995 [A](#)). *shot*³ mutant c4da neurons had shorter dendrites at the L3 stage compared to control MARCM c4da neuron clones (**Fig. 1** [C](#), D). At 16 h APF, most *shot*³ mutant neurons still had dendrites attached to the soma (**Fig. 1** [C'](#) D', G - I). We also generated an sgRNA line targeting a common exon of all Shot splice isoforms and found that it caused significant c4da neuron dendrite pruning defects when expressed in c4da neurons in a sensitized *shot*^{3/+} heterozygous background (**Fig. 1** [E](#) - F', G - I).

Dendrite pruning is the first step of c4da neuron metamorphic remodeling and is followed by the regeneration of new dendrites adapted to the adult stage. We noted that c4da neurons lacking Shot also failed to initiate dendrite regrowth at 72 h APF, when control c4da neurons had regrown long and branched dendrites (Figure S1). We conclude that Shot is broadly required for c4da neuron dendrite remodeling.

Shot is required for uniform dendritic microtubule orientation

We and others have previously shown that c4da neuron dendrite pruning depends on local microtubule disassembly in proximal dendrites (Herzmann et al, 2017 [A](#)). As Shot is a microtubule regulator, we assessed the state of dendritic microtubules by immunofluorescence against the microtubule-associated protein futsch/MAP1B. At 5 h APF, continuous futsch staining could often still be detected in proximal dendrites of *shot* knockdown neurons, while control neurons had large gaps their futsch signal in these regions (Fig. S2). The uniform plus end-in orientation of dendritic microtubules is particularly important for dendrite pruning as microtubules must shrink away from the cell body in a coordinated fashion (Herzmann et al, 2018 [A](#); Rumpf et al, 2019 [A](#), Wang et al, 2019 [A](#)). We therefore asked whether loss of Shot affects dendritic microtubule orientation by using EB1::GFP to monitor the microtubule growth direction. In primary dendrites of third instar control c4da neurons, EB1::GFP comets (the plus ends of growing microtubules) moved exclusively retrogradely towards the soma (**Fig. 2** [A](#), C).

Upon *shot* knockdown, however, a significant proportion of EB1::GFP comets could be seen moving anterogradely (**Fig. 2** [A](#), C). In dendrites of *shot*³ c4da neuron MARCM clones, we observed an even stronger increase in anterogradely moving comets in comparison to control

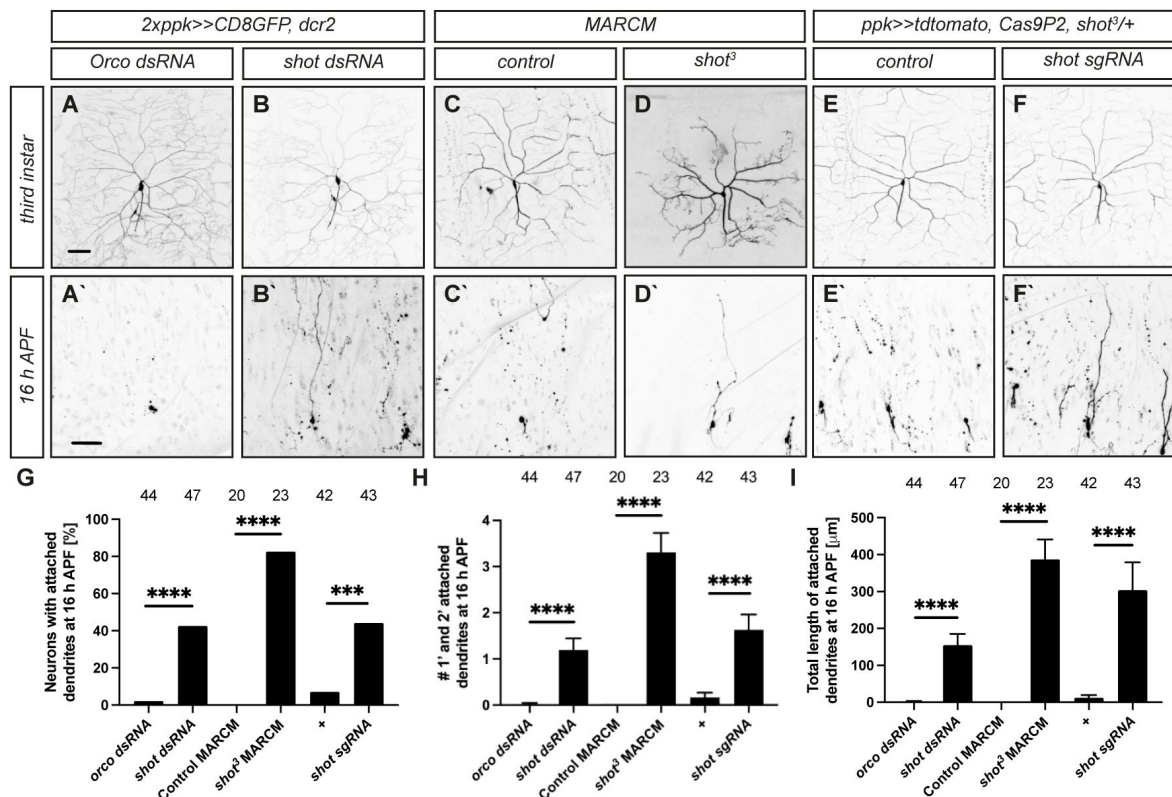


Figure 1.

Shot is required for c4da neuron developmental dendrite pruning.

A – F' C4da neurons of the indicated genotypes were imaged at the third instar larval stage (**A – F**) and at 16 hours after puparium formation (h APF) (**A' – F'**). **A, A'** Control c4da neurons expressing Orco dsRNA under the control of *ppk-GAL4*. **B, B'** C4da neurons expressing Shot dsRNA under the control of *ppk-GAL4*. **C, C'** Control c4da neurons labeled by MARCM. **D, D'** *shot³* mutant c4da neurons labeled by MARCM. **E, E'** Control c4da neurons expressing Cas9P2 under *ppk-GAL4* in a *shot³* heterozygous background. **F, F'** C4da neurons coexpressing Cas9P2 and Shot sgRNA under the control of *ppk-GAL4* in a *shot³* heterozygous background. **G** Phenotypic penetrance of pruning defects in **A' – F'**. Sample sizes are indicated above the graph. *** $P < 0.001$, **** $P < 0.0001$, two-tailed Fisher's exact test. **H** Average number of primary and secondary dendrites attached to the soma at 16 h APF in samples **A' – F'**. Values are mean $\pm$ s. e. m., **** $P < 0.0001$, Mann-Whitney U test. **I** Total length of remaining dendrites at 16 h APF in samples **A' – F'**. Values are mean $\pm$ s. e. m., **** $P < 0.0001$, Mann-Whitney U test. Scale bars in **A** and **A'** are 50 μ m.

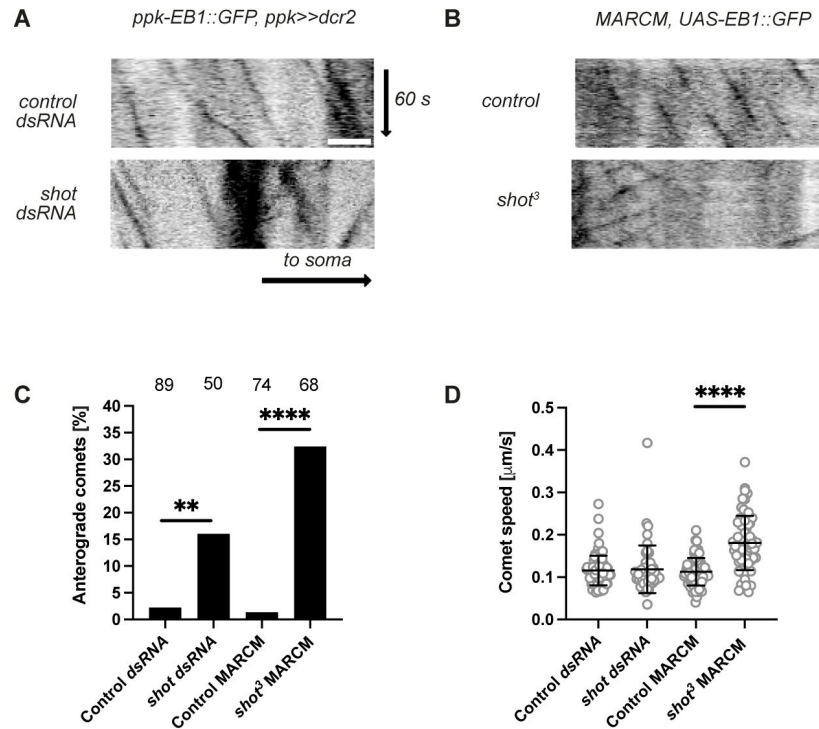


Figure 2.

Shot is required for uniform plus end-in orientation of dendritic microtubules.

A, B EB1::GFP comets were imaged in primary dendrites of third instar larval c4da neurons, and comet movement was depicted in kymographs. Direction of the soma and time are indicated. **A** Upper panel: EB1::GFP comets in control neuron expressing Orco dsRNA; lower panel: EB1::GFP comets in neuron expressing Shot dsRNA. **B** Upper panel: EB1::GFP comets in control c4da neuron MARCM clone; lower panel: EB1::GFP comets in *shot³* mutant c4da neuron MARCM clone. **C** Percentage of anterogradely moving comets in panels **A, B**. N is indicated above the graph. ** $P < 0.01$, **** $P < 0.0001$, two-tailed Fisher's exact test. **D** EB1::GFP comet speed in panels **A, B**. **** $P < 0.0001$, Wilcoxon's test. The scale bar in **A** is 5 μm .

MARCM clones (Fig. 2 B, C). In support of a broad role for Shot in dendritic microtubule organization, EB1 comets in *shot*³ mutant c4da neurons also moved at an increased speed compared to controls (Fig. 2 D). We conclude that Shot is required for the correct orientation of dendritic microtubules, and that this function likely underlies its role in pruning.

Actin and microtubule binding are crucial for Shot function in dendrites

In order to better understand the function of Shot in dendritic pruning and microtubule orientation, we next asked which domains of Shot are crucial for these functions. Of particular interest were the domains of Shot that bind actin and microtubules. Shot has two Calponin Homology (CH) domains at the N-terminus that mediate binding to specific actin structures, and a microtubule-binding GAS2 domain in its C-terminal region (Fig. 3 A). Shot also possesses functionally important binding sites for end binding proteins such as Patronin and EB1. We next expressed full length or truncated Shot variants in *shot*³ mutant c4da neurons using the GAL4/UAS system and asked whether they could rescue the pruning defects of the mutant. A full length Shot UAS transgene (Shot^{FL}) fully rescued the pruning defects of *shot*³ mutant neurons (Fig. 3 B, C, F, G). In contrast, Shot^{βCH1}::GFP, a Shot variant lacking the first CH domain that has decreased affinity for actin (Lee & Kolodziej, 2002), afforded only a partial rescue (Fig. 3 D, F, G). Interestingly, overexpression of a C-terminal Shot fragment containing the microtubule binding domain and adjacent regions, but not the CH or spectrin repeat domains (UAS-Cterm::YFP), also fully rescued the pruning defects of mutant cells (Fig. 3 E - G). In further support of non-functionality, overexpression of Shot^{βCH1} caused strong pruning defects (Fig. S3).

In order to assess the functionality of these Shot variants in dendritic microtubule organization, we made use of kinesin-lacZ (kin::lacZ), a fusion between the kinesin-1 motor domain and β-galactosidase. In control MARCM c4da neurons, kinesin-lacZ localizes exclusively to the soma and axon (Fig. 3 H), as the kinesin motor travels towards microtubule plus ends. In *shot*³ mutant c4da neurons, axonal kin::lacZ was reduced, and it could instead be found in distinct puncta in dendrites, indicating the presence of plus end-out microtubules in these dendrites (Fig. 3 I, M). Re-expression of full length Shot, but not of Shot^{βCH1}::GFP, significantly rescued the number of dendritic kin::lacZ puncta (Fig. 3 J, K, M). As with the pruning defects, overexpression of Shot-Cterm::YFP restored dendritic exclusion of kin::lacZ (Fig. 3 L, M).

It had previously been suggested that Shot and its mammalian homologues can bundle and/or stabilize microtubules. In order to assess the effects of Shot loss on dendritic microtubule structure, we labeled microtubules in c4da neurons by expression of mCherry-tagged α-tubulin and used 2D stimulated emission depletion (STED) microscopy for visualization. In primary dendrites of control c4da neurons, mCherry-tagged microtubules appeared as several long filaments (Fig. S4). In dendrites of c4da neurons lacking Shot, fewer filamentous structures were visible and the staining was more punctate, possibly indicating that these neurons have fewer and/or damaged microtubules (Fig. S4). Thus, both microtubule and actin binding activities of Shot are important for correct dendritic microtubule orientation and pruning.

Shot cooperates with actin to promote microtubule orientation

The above domain analysis demonstrated that the actin binding ability of Shot is important for its function. Spectraplakins often bind to cortical actin structures underneath the plasma membrane. Consistently, we found that endogenously GFP-tagged Shot localized to the rim of the dendrite in third instar larval c4da neuron dendrites (Fig. 4 A, A') in structured illumination microscopy (SIM) imaging. To directly probe for a role for actin in dendritic microtubule organization, we sought to disassemble actin filaments *in vivo*. The actin-severing enzyme Mical is a pruning factor that is induced at the onset of metamorphosis (Kirilly et al, 2009). We used the GAL4/UAS system to overexpress Mical prematurely in larval c4da neurons. In dendrites of c4da neurons overexpressing Mical, a significant fraction of EB1::GFP comets moved anterogradely, indicating

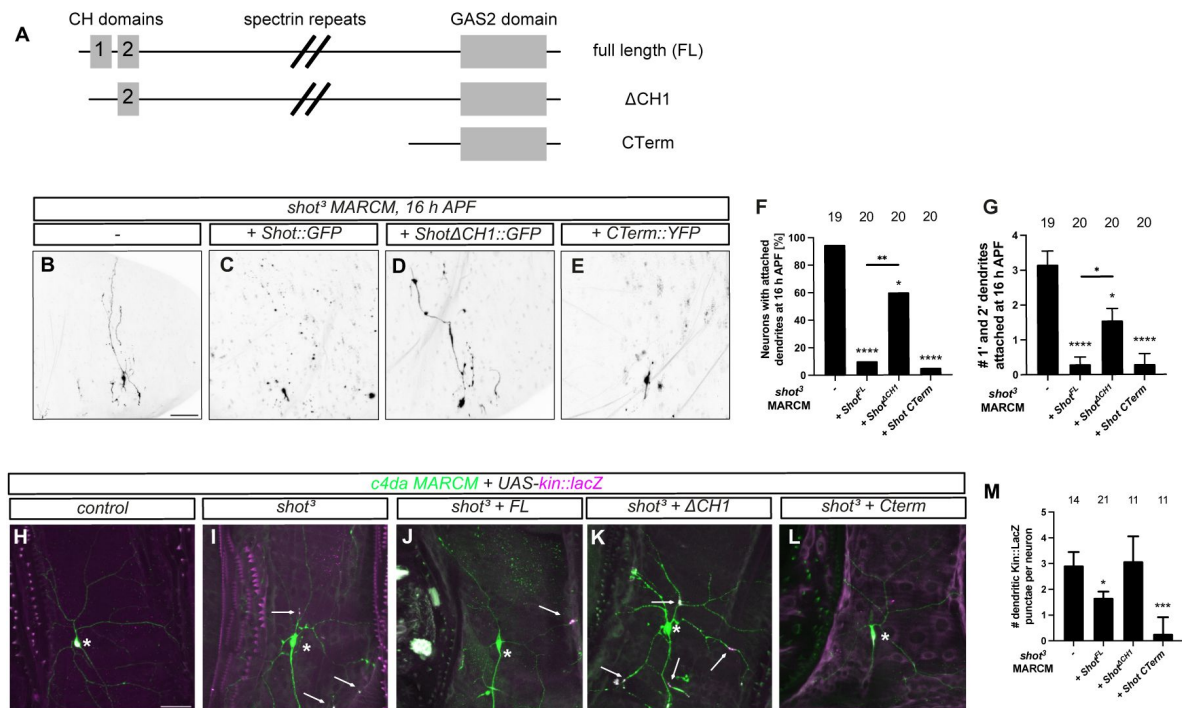


Figure 3.

Shot domains important for pruning and microtubule organization.

A UAS-Shot constructs. CH, Calponin homology. **B - E** Ability of the indicated Shot UAS constructs to rescue the pruning defects of *shot³* mutant c4da neurons at 16 h APF. **B** *shot³* mutant c4da neuron labeled by MARCM. **C** *shot³* mutant c4da neuron expressing full-length Shot::GFP. **D** *shot³* mutant c4da neuron expressing Shot::GFP lacking the CH1 domain. **E** *shot³* mutant c4da neuron expressing Shot C-term::YFP. **F** Penetrance of pruning defects. N is indicated above the graph. * $P < 0.05$, *** $P < 0.001$, **** $P < 0.0001$, two-tailed Fisher's exact test. **G** Number of primary and secondary dendrites attached to the cell body at the indicated timepoints. Values are mean $\pm$ s. e. m., *** $p < 0.001$, **** $p < 0.0001$, Mann-Whitney U test. **H - L** Kin::lacZ localization in third instar c4da neuron MARCM clones. **H** Control c4da neuron. **I** *shot³* mutant c4da neuron. **J** *shot³* mutant c4da neuron expressing full-length Shot. **K** *shot³* mutant c4da neuron expressing Shot lacking the CH1 domain. **L** *shot³* mutant c4da neuron expressing Shot C-term::YFP. Arrows in **H - L** mark dendritic kin::lacZ puncta, asterisks denote soma position. **M** Number of dendritic kin::lacZ puncta in third instar c4da neurons expressing the indicated Shot UAS constructs. N is depicted above the graph. Values are mean $\pm$ s. e. m., * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, Wilcoxon test. Scale bars in **B** and **H** are 50 μ m.

mixed microtubule orientation (Fig. 4 B, C). Mical overexpression combined with Shot knockdown did not lead to an additive increase in the percentage of anterograde comets, indicating epistasis between the manipulations (Fig. 4 B, C). However, these anterograde comets moved significantly faster than their retrograde counterparts (Fig. 4 D). Furthermore, movement of the anterograde comets in this sample was often irregular, and comets frequently slowed down and then sped up again (Fig. 4 B). Such phenomena are often seen when microtubules are not attached to specific anchoring sites such that they can be moved by microtubule motors.

Mical overexpression suppresses the pruning defects of ecdysone signaling mutants (Kirilly et al, 2009; Rode et al, 2018). However, we observed a synergistic enhancement of the pruning defects caused by Shot knockdown when we overexpressed Mical (Fig. 4 E). Thus, Shot's role in dendrite pruning might in part be related to microtubule anchoring.

Shot is recruited to dendritic tips during early dendrite growth

Our analysis of Shot domains showed that the Shot CH1 domain is important for dendrite pruning and microtubule orientation (Fig. 3). This domain is often required to recruit Shot to specific subcellular locations (Nashchekin et al, 2024). To test whether the CH1 domain is required for localization to the dendrite membrane (Fig. 4 A), we visualized transgenic Shot or Shot^{ΔCH1} in third instar c4da neurons using SIM. Similar to endogenously GFP-tagged Shot, transgenic full length Shot::GFP localized in puncta and sometimes streaks (especially around dendritic branchpoints) close to the plasma membrane (Fig. 5 A). Shot^{ΔCH1}::GFP localized in a similar pattern to full-length Shot (Fig. 5 B), indicating that the CH1 domain is unlikely to be required for Shot cortical localization in dendritic shafts.

Since dendritic microtubule organization is set up early during development, we investigated the localization of transgenic Shot::GFP in c4da neurons at the first larval instar stage, when c4da neuron dendrites are growing very fast. At this stage, Shot::GFP was highly enriched at dendrite tips, but could only be detected at low levels in the soma and the more proximal dendritic shafts (Fig. 5 C, E). Shot tip enrichment seemed transient, as it was not observed in mature c4da neuron dendrites at the third instar (Fig. S5). In contrast to full length Shot, Shot^{ΔCH1}::GFP showed significantly lower tip occupancy (Fig. 5 D, F) and often localized in dots that could be several micrometers away from the tips (Fig. 5 D, E). To independently test whether Shot tip localization was actin-dependent, we used UAS-Mical for enzymatic actin disruption. Mical overexpression did not abolish Shot at tips, but caused a significant broadening of the tip signal, again indicating disturbed Shot recruitment (Fig. 5 G, H). Thus, efficient dendrite tip recruitment of Shot during the early stages of c4da neuron dendrite growth requires the CH1 domain.

Rab11 is a component of a developmental dendritic MTOC

The fact that Shot variants lacking the CH1 domain cannot rescue the pruning defects of *shot*³ mutants suggested that dendrite tip localization of Shot was important for its function. The tips of growing dendrites have recently been linked to microtubule organization in *C. elegans* PVD sensory neurons. Here, tip-localized Rab11-positive vesicles recruit a ncMTOC (Liang et al, 2020; Harterink et al, 2018). The composition and localization of dendritic MTOCs in *Drosophila* sensory (c4da and c1da) neurons are debated. Microtubules have been observed originating in dendritic shafts, at branchpoints and at tips, and both Golgi outposts and Rab5 endosomes have been implicated as sites harbouring MTOCs (Ori-McKenney et al, 2012; Nguyen et al, 2014; Mukherjee et al, 2020; Yalgin et al, 2015; Weiner et al, 2020). Most studies in c4da neurons have focused on mature dendrites in third instar neurons. We therefore asked whether the same factors and/or a similar mechanism as in PVD neurons also act in early developing c4da neurons.

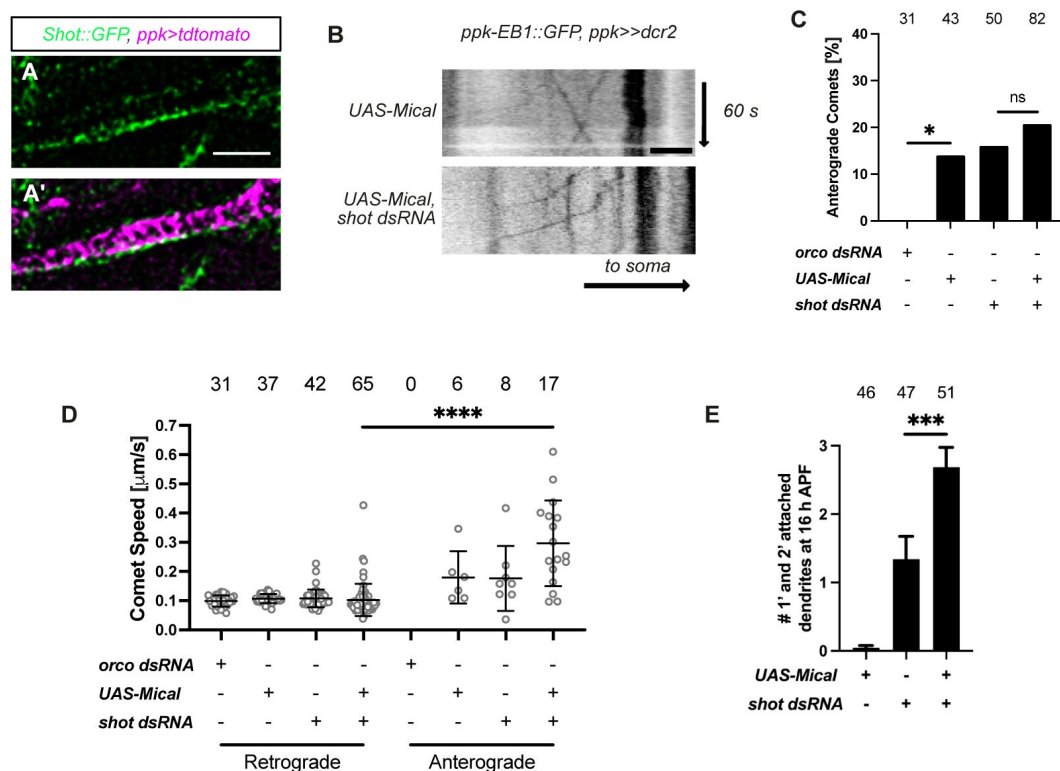


Figure 4.

Shot links microtubule regulation to actin.

A, A' Localization of Shot in a third instar primary c4da neuron dendrite. Endogenously tagged Shot::GFP was labeled by immunofluorescence and visualized by structured illumination microscopy. C4da neurons were labeled by expression of cytosolic tdtomato under *ppk-GAL4*. **A** Shot::GFP. **A'** merge with tdtomato. **B** Effect of actin severing on EB1::GFP comet movement. Mical was overexpressed in c4da neurons with or without Shot knockdown, and EB1 comets were analyzed as in Fig. 2. **B**. Upper panel, kymograph of EB1 comet movement in c4da neuron overexpressing Mical. Lower panel, kymograph of EB1 comet movement upon Mical overexpression and *shot* knockdown. **C** Penetration of anterograde comets in Fig. 4. **B**. N is given in the graph. * $P < 0.05$, Fisher's exact test. **D** Graph depicting the effects of Shot knockdown and Mical expression on EB1 comet speed. N is given in the graph. **** $P < 0.0001$, Wilcoxon's test. **E** Synergistic effects of actin severing and Shot knockdown on c4da neuron dendrite pruning. Penetration of pruning defects at 16 h APF in c4da neurons overexpressing Mical, upon Shot knockdown, or both. N is given in the graph. * $P < 0.05$, Fisher's exact test. Scale bars in **A** and **B** are 5 μm .

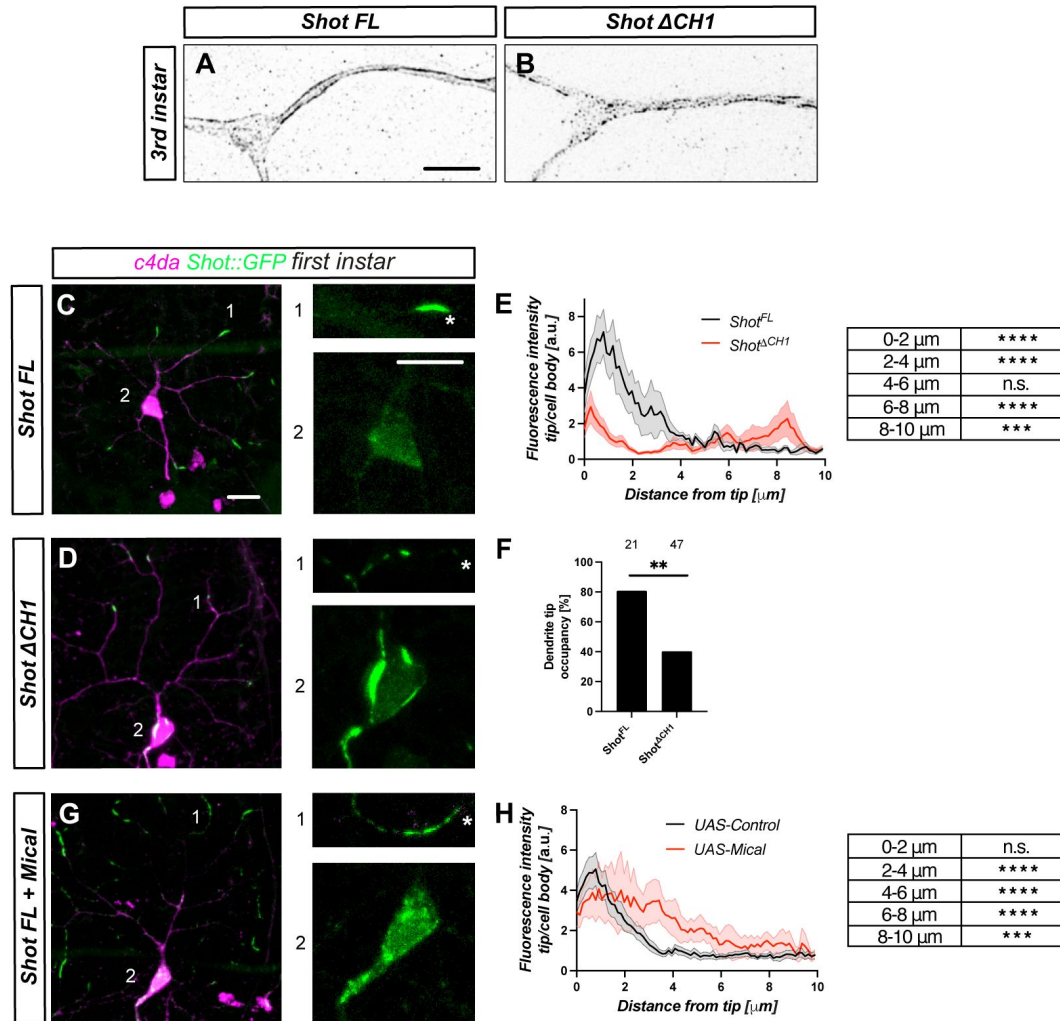


Figure 5.

Shot is recruited to tips of growing dendrites via its CH1 domain.

The indicated GFP-tagged Shot constructs and tdtomato were expressed in *c4da* neurons under *ppk-GAL4* and visualized at the indicated developmental stages. **A, B** Dendritic shaft localization of transgenic Shot::GFP variants in third instar *c4da* neuron dendrites was visualized by immunofluorescence and structured illumination microscopy (SIM). **A** Localization of full-length Shot (Shot^{FL}::GFP). **B** Localization of Shot lacking the CH1 domain (Shot ^{Δ CH1}::GFP). **C - H** Localization of Shot variants in first instar *c4da* neurons. Images on the right show close-ups of dendrite tip (1) and soma regions (2). The asterisk in the dendrite close-up marks the position of the tip. **C** Localization of Shot^{FL}::GFP. **D** Shot ^{Δ CH1}::GFP. **E** Fluorescence intensity profiles of Shot::GFP and Shot ^{Δ CH1}::GFP in the distal 10 μ m of first instar dendrites. Intensity values were normalized to Shot::GFP intensity in the cell body. Solid lines indicate average, envelopes indicate S. D. (N=10). The table shows significance between genotypes for the indicated distances from the tip **** P<0.0001, *** P<0.001, n. s. not significant, Mann Whitney U test. **F** Dendrite tip occupancy of Shot^{FL}::GFP and Shot ^{Δ CH1}::GFP. ** P<0.01, Fisher's exact test. **G** Effect of Mical overexpression on Shot^{FL}::GFP localization. Images on the right show close-ups of dendrite tip (1) and soma regions (2). **H** Fluorescence intensity profiles of Shot^{FL}::GFP in the distal 10 μ m of first instar dendrites with or without Mical overexpression. Quantification was as in **E**. N=10, **** P<0.0001, *** P<0.001, ** P<0.01, n. s. not significant, Mann Whitney U test. Scale bars are 5 μ m in **A** and 10 μ m in **C** (larger image and close-up).

Rab11 is required for c4da neuron dendrite pruning (Krämer et al, 2019 [DOI](#); Lin et al, 2020 [DOI](#)). Given the strong dependence of pruning on microtubule organization, it is interesting to speculate that it might perform a microtubule-related function in early c4da neuron dendrites as well. We first tested whether Rab11 colocalizes with the bona fide MTOC component γ -tubulin. To this end, we coexpressed Rab11::mCherry with GFP-tagged γ -tubulin in c4da neurons. At the first instar, a subset of Rab11::mCherry puncta - likely vesicles - could be seen overlapping with γ -tubulin::GFP in discrete dendritic puncta that could also be found at dendrite tips (Fig. 6 [DOI](#) A). To test whether such puncta could nucleate microtubules, we observed Rab11::mCherry together with EB1::GFP. In first instar neurons, EB1::GFP comets could be observed originating from dendritic Rab11::mCherry puncta (6/9 neurons, 10/58 comets) (Fig. 6 [DOI](#) B, C). These comets were mostly seen in dendritic shafts or at branchpoints, likely because we could only capture a subset of EB1::GFP comets in our experiments. No EB1::GFP comets could be observed from Rab11::mCherry puncta at the third instar (0/9 neurons, 27 comets) (Fig. 6 [DOI](#) C). To test for a functional role for Rab11 in dendritic microtubule organization, we knocked it down and visualized growing microtubules at the third instar stage. Rab11 knockdown caused a significant fraction of EB1 comets (16 %) to move anterogradely, and overall comet speed was increased (Fig. 6 [DOI](#) D - F). In contrast, a dsRNA targeting Rab5 did not cause a significant increase in anterograde comets in c4da neurons, and comet speed was normal (Fig. 6 [DOI](#) D - F).

The microtubule polymerase Msps/XMAP215 is another potential MTOC component linked to pruning and dendritic microtubule organization (Tang et al, 2020 [DOI](#)). In co-immunoprecipitations from transfected S2 cells, GFP-tagged Rab11, but not GTP-binding-deficient Rab11 S25N, could be co-precipitated with FLAG-tagged Msps (Fig. 6 [DOI](#) G). Lastly, the minus end-binder Patronin is required for dendrite pruning and localizes to growing dendrite tips in PVD neurons (Wang et al, 2019 [DOI](#); He et al, 2020 [DOI](#)). Transgenic Patronin::GFP was also enriched at dendrite tips - as well as in streaks along dendrite shafts - in first instar c4da neurons (Fig. 6 [DOI](#) H). Taken together, our data suggest the existence of a developmentally regulated, Rab11-based MTOC in early growing c4da neuron dendrites that likely also acts at tips.

Shot recruits EB1 to tips and interacts with candidate MTOC components

The above observations opened up the possibility that Shot is part of an early developmental dendritic tip MTOC. In support of this notion, loss of Shot already caused altered microtubule orientation at the first instar stage (Fig. 7 [DOI](#) A). To test whether Shot can recruit the microtubule nucleation machinery, we coexpressed Shot::GFP with red fluorescently tagged EB1 in c4da neurons. In the presence of Shot::GFP, EB1 also became recruited to dendrite tips at the first instar and colocalized with Shot::GFP (Fig. 7 [DOI](#) B, C), furthermore, EB1 comets could be seen emanating from these Shot/EB1-rich tips (Fig. 7 [DOI](#) D). To link Shot to the other microtubule regulators implicated in early dendritic microtubule nucleation, we tested for genetic interactions using pruning as a readout. To this end, we knocked down *shot* in c4da neurons and then crossed in heterozygous P element mutants in *Patronin* and *Msps*, or a dsRNA construct targeting EB1. While these manipulations on their own did not cause pruning defects, the number of secondary and higher order dendrite branches attached to the soma upon was strongly increased when combined with *shot* knockdown, indicating phenotypic enhancement (Fig. 7 [DOI](#) E - I). To assess genetic interactions with Rab11, we knocked down *rab11*, which by itself already causes highly penetrant pruning defects, but most higher order dendrites are still pruned and only primary dendrites remain (Krämer et al, 2019 [DOI](#)) (Fig. 7 [DOI](#) J, L). In contrast, *rab11* knockdown in a *shot*^{3/+} heterozygous background also caused significantly more higher order dendrites to stay attached to the soma than with *rab11* knockdown alone (Fig. 7 [DOI](#) K, L). Thus, EB1 imaging and genetic interactions link Shot to dendritic MTOC functions.

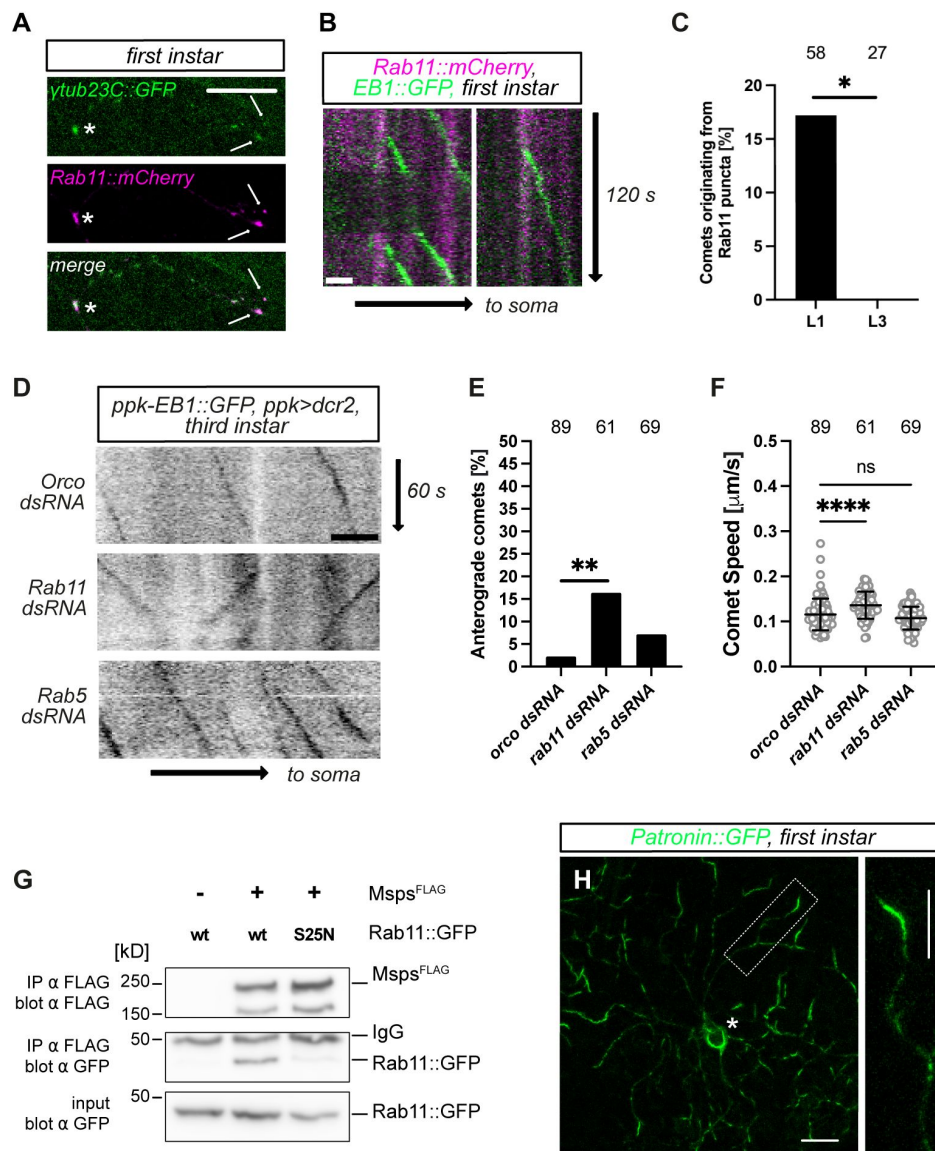


Figure 6.

Evidence that Rab11 is part of a developmentally regulated dendritic microtubule organizing center.

A Rab11 colocalizes with the MTOC marker γ -tubulin in growing c4da neuron dendrites. Rab11::mCherry and γ -tubulin23C::GFP were coexpressed under *ppk-GAL4* and visualized at the first instar. Arrows show Rab11/ γ -tubulin-positive puncta at dendrite tips, the asterisk denotes a double-positive dot at a branchpoint. **B** Microtubules can be nucleated at Rab11 puncta in dendrites. Rab11::mCherry and EB1::GFP were coexpressed in c4da neurons, and EB1 comets were visualized at the first instar. Example kymographs show Rab11::mCherry (magenta) and EB1::GFP (green). **C** Percentage of EB1 comets arising from dendritic Rab11 puncta at the first and third instars. * $P < 0.05$, Fisher's exact test. **D** Effect of Rab11 knockdown on dendritic microtubule orientation. Kymographs show EB1::GFP movement in third instar c4da neuron dendrites. Upper panel, control c4da neuron expressing Orco dsRNA; middle panel, c4da neuron expressing Rab11 dsRNA; lower panel, c4da neuron expressing Rab5 dsRNA. **E** Penetrance of anterograde comets in **D**. ** $P < 0.01$, two-tailed Fisher's exact's test. **F** Speed of EB1::GFP comets in **D**. **** $P < 0.0001$, Wilcoxon's test. **G** GFP-tagged Rab11 (wt or S25N) and FLAG-tagged Msp were cotransfected into S2 cells and immunoprecipitated with FLAG beads. Inputs and immunoprecipitates (IP) were blotted with the indicated antibodies. Sizes of molecular weight markers in kiloDalton (kD) are shown. IgG denotes antibody heavy chains. **H** Patronin localizes along dendrites and in dendritic tips in first instar c4da neurons. The side panel shows an enlarged image of a dendrite (boxed area in larger image). The asterisk denotes the position of the soma. Scale bars are 5 μm in **A** and **D**, 2 μm in **B** and 10 μm in **G**.

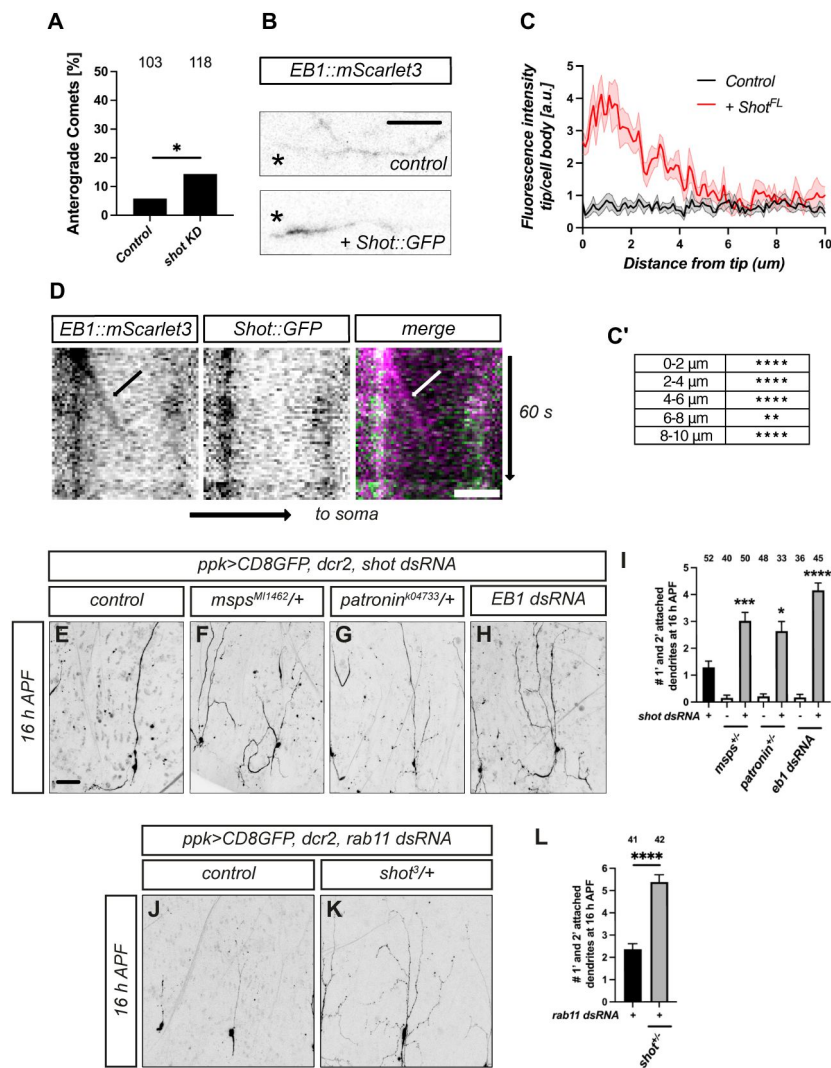


Figure 7.

Evidence that Shot acts as part of a dendritic MTOC.

A Frequency of anterograde EB1 comets at the L1 stage in control (Orco dsRNA) and shot knockdown neurons (shot dsRNA, *shot*^{3/+}). * $P < 0.05$, Mann Whitney U test. **B** Shot recruits EB1 to dendrite tips. EB1::mScarlet3 was expressed in c4da neurons without (upper panel) or with Shot^{FL}::GFP (lower panel) and visualized at the first instar larval stage. The position of the dendrite tip is indicated by an asterisk. **C** Fluorescence intensity profiles of EB1::mScarlet3 in the distal 10 μm of first instar dendrites. Solid lines indicate average, envelopes indicate S. D. (N=9 each). **C'** Table showing significance between genotypes for the indicated distances from the tip. **** $P < 0.0001$, ** $P < 0.01$, Mann Whitney U test. **D** Example kymograph of EB1::mScarlet3 and Shot::GFP at a dendritic tip. Tip position is to the left, and arrows in the EB1::mScarlet3 and merge panels indicate a comet originating there. **E - L** Synergistic genetic interactions between *shot* and other dendritic microtubule orientation/MTOC factors during c4da neuron dendrite pruning. **E - I** *shot* dsRNA was expressed in c4da neurons of animals in the indicated backgrounds, and pruning defects were quantified at 16 h APF. **E** C4da neuron expressing *shot* dsRNA. **F** C4da neuron expressing *shot* dsRNA in a *msps*^{+/+} background. **G** C4da neuron expressing *shot* dsRNA in a *patronin*^{+/+} background. **H** C4da neuron co-expressing *shot* and *EB1* dsRNAs. **I** Severity of pruning defects in **E - G**. N = 33 - 52, ** $P < 0.01$, *** $P < 0.001$, Wilcoxon's test. **J, K** *rab11* dsRNA was expressed in c4da neurons of control animals (**J**) or in a *shot*^{3/+} heterozygous background (**K**), and pruning defects were quantified at 16 h APF. **L** Severity of pruning defects in **J, K**. N = 41 and 42. **** $P < 0.0001$, Wilcoxon's test. Scale bars are 5 μm in **B**, 2 μm in **D** and 50 μm in **E**.

Discussion

Microtubule organization in neurites is crucial for neuronal polarity, but the developmental mechanisms are still emerging. Here, we show that the spectraplakins Shot is required for the uniform plus end-in organization of dendritic microtubules. Our data indicate that the role of Shot is twofold. On the one hand, we provide evidence that Shot anchors microtubules via actin in mature neurons. If unanchored, a subset of plus end-out oriented microtubules in Shot loss-of-function neurons can undergo sliding, which has been linked to microtubule misorientation (He et al, 2020 [DOI](#)). The observation that only plus end-out microtubules show this behavior suggests that there might be other, orientation-dependent anchoring mechanisms that do not depend on Shot. Surprisingly, overexpression of the GAS2 microtubule binding domain was sufficient to rescue the phenotypes of *shot* mutant neurons. Microtubule length regulation and catastrophe rescue have recently been shown to be important for maintenance of dendritic microtubule orientation (Liang et al, 2024 [DOI](#)). It will be interesting to test whether the GAS2 domain can fulfil a similar function in c4da neurons.

The other function of Shot is as part of an early dendritic MTOC. Similar to the *C. elegans* PVD neuron, c4da neurons have a transient microtubule-linked structure at the tip of early growing dendrites, and they depend on Rab11 for proper microtubule organization. Shot interacts genetically with Rab11 and is a component of this tip structure. Early-stage tip MTOCs and Rab11 dependence may therefore be a general feature of dendritic development. The exact function of Shot in the tip MTOC is still unclear, but the observation that Shot can recruit EB1 to tips suggests that it may recruit several MTOC components. The dendritic tip MTOCs of *C. elegans* were suggested to be transported to dendrite tips on transient plus end-out microtubules (Liang et al, 2020 [DOI](#)). Plus end-out microtubules can be observed in terminal c4da neuron dendrites (Ori-McKenney et al, 2012 [DOI](#); Hu et al, 2022 [DOI](#)). It is interesting to speculate that Shot might participate in such a transport function. Crucially, both proposed functions of Shot during microtubule organization depend on actin binding, linking the two main cytoskeletal components in neurite organization.

Materials and methods

Fly Strains

C4da neurons were labeled by UAS-CD8::GFP or UAS-tdtomato expression under *ppk-GAL4* (Grueber et al, 2007 [DOI](#)). MARCM clones of *shot*³ mutants (BL 5141) were induced with *SOP-FLP* (Matsubara et al, 2011 [DOI](#)) and labeled by tdtomato expression under *nsyb-GAL4*^{R57C10}. Endogenously tagged Shot::GFP was from (Voelzmann et al, 2024 [DOI](#)). EB1 analyses were carried out using either UAS-EB1::GFP (Zheng et al, 2008 [DOI](#)), *ppk-EB1::GFP* (Arthur et al, 2015 [DOI](#)), or UAS-EB1::mScarlet3. The following dsRNA lines were used: Orco (BL 31278) as control, Shot (BL 28336), Rab11 (VDRC 22198), Rab5 (VDRC 34096), EB1 (VDRC 24451). All dsRNAs were coexpressed with UAS-dcr2 (Dietzl et al, 2007 [DOI](#)). UAS transgenes were UAS-Shot^{RE}::GFP on the X chromosome (BL 29044), or mobilized to the third chromosome (this study), UAS-Shot^{RC}::GFP (1CH1, BL 29042), UAS-Shot Cterm::YFP (Nashchekin et al, 2024 [DOI](#)), UAS-kin::lacZ (Clark et al, 1997 [DOI](#)), UAS-Mical (Terman et al, 2002 [DOI](#)), UAS-*atub84B*::mCherry (Villars et al, 2022 [DOI](#)), UAS-Rab11::GFP (BL 8506), UAS-*tau*^{HA} (Herzmann et al, 2017 [DOI](#)), UAS-*ytub23C*::GFP (Nguyen et al, 2011 [DOI](#)), UAS-Patronin::GFP (Derivery et al, 2015 [DOI](#)), UAS- *cytoABKAR* TA (Marzano et al, 2021 [DOI](#)) as inert UAS control. Additional mutant alleles were *mpps*^{MI14162} (BL 59478), *patronin*^{k07433} (Kyoto 111217).

Cloning and transgenes

Shot sgRNAs GCTGCCCTCTCAGGCCGATT (target 1) and GAGTTCTCCAGAGTGGTCAC (target 2) were cloned into pCFD4w+ (gift from S. Schirmeier). For transgenesis, plasmids were injected into flies carrying the 86Fb acceptor site. C-terminally GFP-tagged Rab11 for biochemistry was cloned into pUAST attB carrying a C-terminal GFP tag using TOPO/TA cloning. The S25N mutation was inserted by PCR mutagenesis. For Rab11::mCherry, the Rab11 entry clone was used to generate pUAST attB Rab11::mCherry, the insert was subcloned into pUAST, which was then injected into *w¹¹¹⁸* flies using classical transposase-mediated transgenesis. To generate EB1::mScarlet3, a *Drosophila* codon optimized mScarlet3 (Gaudella et al, 2023 [DOI](#), synthesized by Twist Bioscience) was added after the EB1 coding sequence separated by a CACACCTCCACTACCGCTAGCAGGCCGCCACGCGTGGTACCTTCTGGTCCA linker and cloned into pUAST. The resulting plasmid was injected into wild type embryos by Bestgene. C-terminally FLAG-tagged Msps (isoform RB) was cloned into pUAST attB using standard cloning procedures.

Immunofluorescence

Briefly, appropriately staged pupal filets were fixed in 4% formaldehyde, blocked in PBS with 0.3 % Triton X-100 and 10% goat serum, and incubated with antibody in blocking buffer over night. C4da neurons labeled by *ppk* promoter fusions, endogenous or transgenic GFP-tagged Shot were visualized with chicken (1:500, Aves labs), rabbit (1:1000, Invitrogen A11122) or mouse (1:1000, Invitrogen A1120) anti-GFP. Tdtomato and mcherry-tagged transgenes were detected with rat anti-mCherry (1:1000, Invitrogen) or rabbit anti-DsRed antibodies (1:1000, Clontech). Kin:: β -galactosidase fusion proteins were detected with rabbit anti- β -galactosidase (Cappel, preabsorbed in-house, 1:700). Secondary antibodies for regular immunofluorescence were conjugated to Alexa Fluor 488, 568 or 647. For STED microscopy of tagged tubulin, Abberior anti rabbit STAR RED were used as secondary antibodies.

Microscopy, time lapse imaging

For analyses of dendrite pruning phenotypes at 16 h APF, animals were dissected out of the pupal case and dorsal ddaC c4da neurons in segments A2 - A5 were imaged live on a Zeiss LSM710 confocal microscope with a 20x Plan Apochromat water objective (1.0 NA). For regrowth analyses at 72 h APF, animals were dissected out of the pupal case and ventral v'ada c4da neurons in segments A2 - A4 were imaged as described (Sanal et al, 2023 [DOI](#)). For live protein localization analyses, first instar larvae were immobilized on double sided tape, third instar larvae were briefly anaesthetized with ether before imaging. Neurons were then imaged live on Zeiss LSM710 or LSM880 microscopes with 63x Plan Apochromat DIC M27 oil objectives (1.4 NA). EB1::GFP imaging was performed on a Zeiss LSM880 microscope using a 40x Plan Apochromat FCS M27 (1.2 NA) oil objective. Consecutive images of a single plane were taken every second for 1 - 2 minutes.

Immunofluorescence images of kin::lacZ were obtained on a Zeiss LSM 710 microscope using a 40x Zeiss C Apochromat water objective (1.1 NA). Structured illumination microscopy (SIM) was performed on a Zeiss Elyra 7 microscope with a 63x Plan Apochromat DIC M27 oil objective (1.4 NA) with an Edge 4.2 SIM camera (50 ms exposure time), and deconvolution was done using SIM² software. All microscopes used ZEN Black software. Stimulated Emission Depletion Microscopy (STED) images were taken with an abberior STEDYCON equipped with an inverted IX83 microscope (Olympus), a 100x oil objective (Olympus UPLXAPO100XO, NA 1.45), using pulsed excitation lasers at 640 nm and a pulsed STED laser operating at 775 nm, continuous autofocus and gated detection with avalanche photodiode element detectors. All acquisition operations were controlled by STEDYCON Software (abberior Instruments), and images were deconvolved using Huygens deconvolution software. Microscopic images shown are maximum projections or (where

indicated) single plane images. All processing was done in Fiji (Schindelin et al, 2012 [↗](#)) using the plug-ins NeuronJ for dendrite length measurements, and Image Stabilizer for EB1::GFP comet analysis. Kymographs from different channels were overlaid using the KymoResliceWide plugin.

S2 cell culture, immunoprecipitation and Western blots

pUAST Rab11::GFP (wt or S25N) was cotransfected into S2 cells with pUAST- Msp^{FLAG} and Actin5C-GAL4 as described (Herzmann et al, 2017 [↗](#)). After 72 hours, cells were harvested in ice-cold PBS, and lysed in cold lysis buffer (100 mM NaCl, 5 mM MgCl₂, 50 mM Tris/HCl pH 7.6, 5 % glycerol, 1 % Triton X-100, 1x complete protease inhibitor). Cleared lysates were incubated with anti-FLAG beads (Sigma A- 2220) for 2 hours. After three washes with lysis buffer, bound proteins were eluted in SDS sample buffer. Samples were run on 8 % gels and blotted with JL-8 anti-GFP (Clontech, 1:1000) and FLAG M2 (Sigma F3165, 1:5000) antibodies. Bands were visualized on an Amersham Imager 680.

Quantification and statistical analyses

Phenotypic penetrance was assessed by counting the number of neurons with dendrites still attached to the soma. Here, significance was determined using a categorical two-tailed Fisher's exact test (graphpad.com). Length of unpruned dendrites were measured using the Fiji NeuronJ plugin and compared using the Wilcoxon Mann Whitney test (Marx et al., 2016).

In order to quantify Shot dendrite tip localization, intensity profiles of the distal 10 µm of each dendrite were averaged. The resulting curves were then sectioned into 2 µm segments for statistical comparisons between genotypes using the Mann Whitney test.

Supplementary figures

Figure S1

(related to Fig. 1). Loss of Shot, patronin and EB1 causes c4da neuron dendrite regrowth defects.

A - D The indicated microtubule regulators were knocked down in c4da neurons under *ppk-GAL4*, and neurons were imaged at 72 h APF. **A** Control c4da neuron expressing Orco dsRNA (N = 22). **B** C4da neuron expressing patronin dsRNA (N = 12). **C** C4da neuron expressing Shot dsRNA (N = 10). **D** C4da neuron expressing EB1 dsRNA (N = 10). **E** Total dendrite length in **A - D**. Values are mean $\pm$ s. d., *** $P < 0.001$, **** $P < 0.0001$, Wilcoxon's test. The scale bar in **A** is 50 μ m.

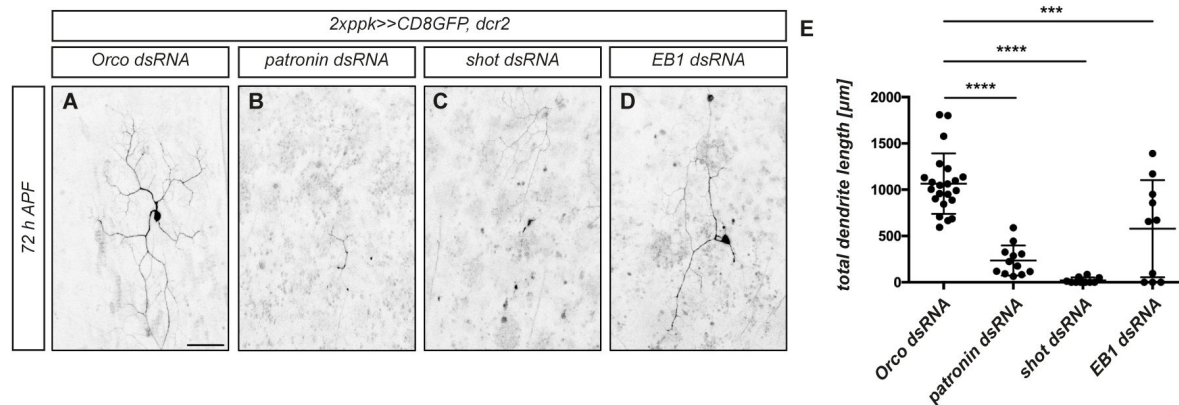


Figure S2

(related to Fig. 2). Loss of Shot delays dendritic microtubule disassembly during dendrite pruning.

A - B Microtubules in early pupal control or Shot knockdown c4da neurons (green) were visualized by futsch/22C10 staining (magenta) at 6 h APF. **A** Control c4da neuron. **A'** futsch/22C10 signal in boxed area in **A**. The arrow indicates a c4da neuron dendrite. **B** C4da neuron expressing Shot dsRNA. **B'** futsch/22C10 signal in boxed area in **B**. The arrow indicates a c4da neuron dendrite. **C** Graph depicting the average number of neurons with continuous futsch/22C10 staining. Data are mean $\pm$ s. d., N = 10 each, ** $P < 0.01$, student's t- test. The scale bar in **A** is 20 μ m.

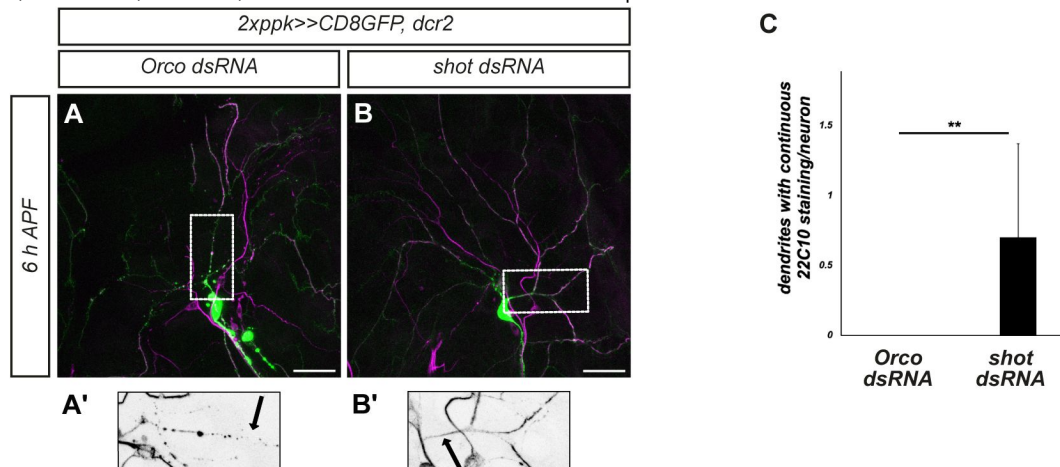


Figure S3

(related to Fig. 3). Effect of Shot overexpression on dendrite pruning.

A - E Full length or truncated Shot variants were overexpressed in c4da neurons, and the effects on dendrite pruning were assessed at 16 h APF. **A** Control c4da neuron (N = 30). **B** C4da neuron overexpressing full length Shot::GFP (N = 42). **C** C4da neuron overexpressing Shot^{ΔCH1}::GFP domain (N = 54). **D** C4da neuron overexpressing Shot Cterm::YFP, containing the microtubule binding domain of Shot (N = 54). **E** Penetrance of pruning defects in A - D. * P<0.05, **** P<0.0001, Fisher's exact test. **F** Severity of pruning defects in A - D. Values are mean +/- s. e. m., * P<0.05, **** P<0.0001, Wilcoxon's test. The scale bar in A is 50 μm.

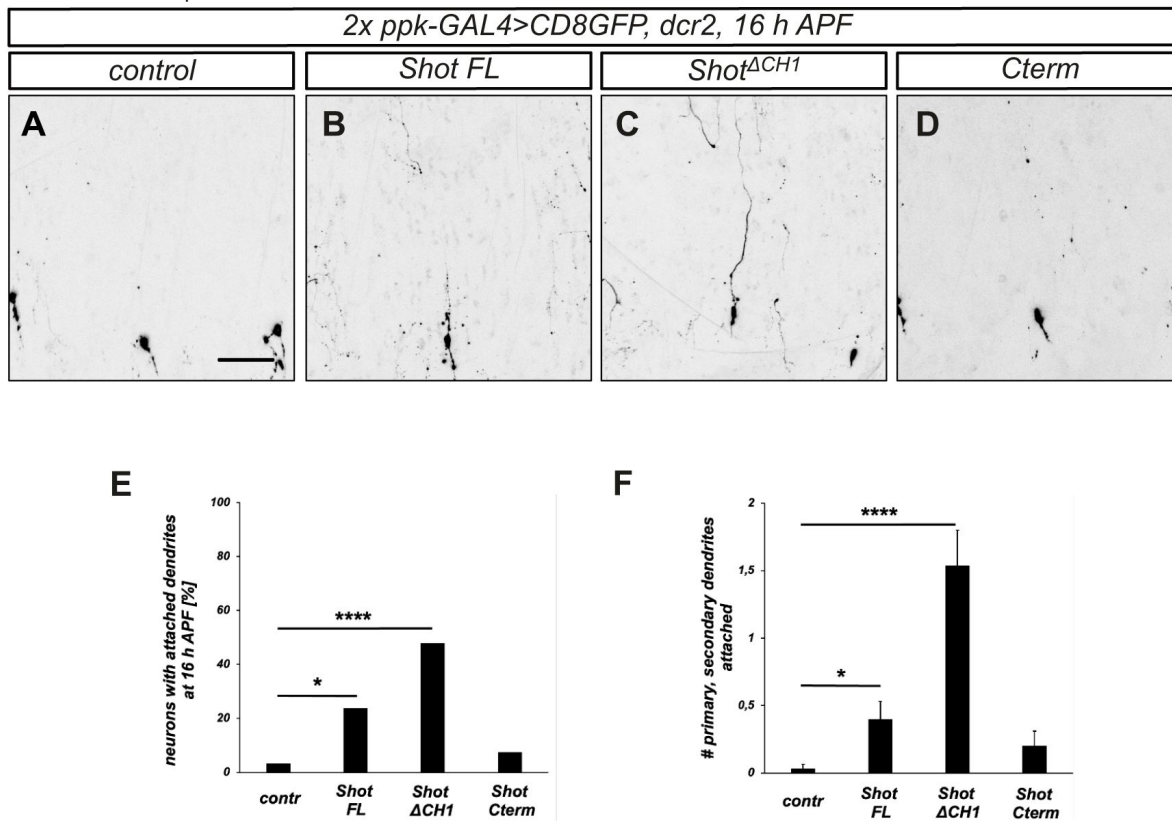
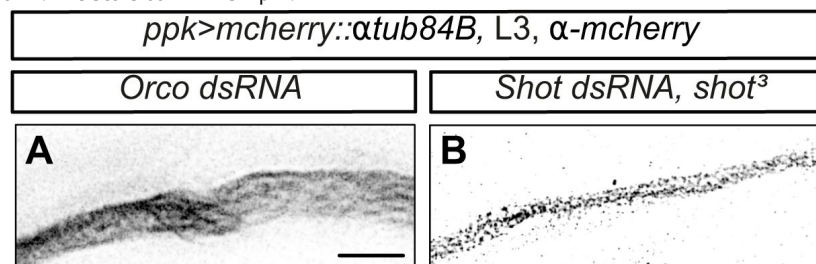


Figure S4

(related to Fig. 3). Effect of Shot knockdown on dendritic microtubule structure.

Microtubules were visualized by immunofluorescence of mCherry::α-tubulin (under ppk-GAL4) followed by 2D STED. Images show primary dendrites. **A** Microtubules in control c4da neuron expressing a control dsRNA against Orco. **B** Microtubules upon Shot knockdown. The scale bar in A is 2 μm.



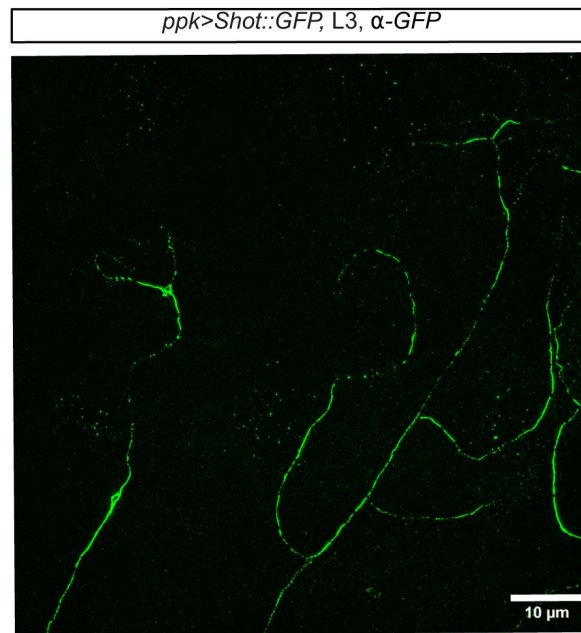


Figure S5

(related to Fig. 5). Shot is not enriched at dendritic tips in mature neurons.

Shot^{FL}::GFP was expressed in c4da neurons under *ppk-GAL4* and a region comprising both dendrite shafts and tips of a third instar c4da neuron was visualized by immunofluorescence and structured illumination microscopy (SIM). The scale bar is 10 μm.

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Additional information

Author Contributions

M. Davies designed experiments, performed phenotypic analyses and contributed reagents. N. Sanal and N. Wolterhoff performed phenotypic analyses. U. Gigengack performed biochemical experiments and contributed reagents. Y. Shen and I. Hahn contributed reagents. S. Rumpf designed experiments, performed biochemical analyses and contributed reagents. S. Rumpf wrote the manuscript with input from all other authors.

References

- Akhmanova A, Kapitein LC (2022) **Mechanisms of microtubule organization in differentiated animal cells** *Nat Rev Mol Cell Biol* **23**:541–558 [Google Scholar](#)
- Alves-Silva J, Sánchez-Soriano N, Beaven R, Klein M, Parkin J, Millard TH, et al (2012) **Spectraplakins promote microtubule-mediated axonal growth by functioning as structural microtubule-associated proteins and EB1-dependent +TIPs (tip interacting proteins)** *J Neurosci* **32**:9143–9158 [Google Scholar](#)
- Arthur AL, Yang SZ, Abellana AM, Wildonger J (2015) **Dendrite arborization requires the dynein cofactor NudE** *J Cell Sci* **128**:2191–2201 [Google Scholar](#)
- Clark IE, Jan LY, Jan YN (1997) **Reciprocal localization of Nod and kinesin fusion proteins indicates microtubule polarity in the Drosophila oocyte, epithelium, neuron and muscle** *Development* **124**:461–470 [Google Scholar](#)
- Cox DN, Lu B, Sun TQ, Williams LT, Jan YN (2001) **Drosophila par-1 is required for oocyte differentiation and microtubule organization** *Curr Biol* **11**:75–87 [Google Scholar](#)
- Derivery E, Seum C, Daeden A, Loubéry S, Holtzer L, Jülicher F, et al (2015) **Polarized endosome dynamics by spindle asymmetry during asymmetric cell division** *Nature* **528**:280–285 [Google Scholar](#)
- Dietzl G, Chen D, Schnorrer F, Su K-C, Barinova Y, Fellner M, et al (2007) **A genome-wide transgenic RNAi library for conditional gene inactivation in Drosophila** *Nature* **448**:151–156 [Google Scholar](#)
- Furusawa K, Emoto K (2021) **Spatiotemporal regulation of developmental neurite pruning: Molecular and cellular insights from Drosophila models** *Neurosci Res* **167**:54–63 [Google Scholar](#)
- Gadella TWJ, van Weeren L, Stouthamer J, Hink MA, Wolters AHG, Giepmans BNG, et al (2023) **mScarlet3: a brilliant and fast-maturing red fluorescent protein** *Nat Methods* **20**:541–545 [Google Scholar](#)
- Grueber WB, Ye B, Yang C-H, Younger S, Borden K, Jan LY, et al (2007) **Projections of Drosophila multidendritic neurons in the central nervous system: links with peripheral dendrite morphology** *Development* **134**:55–64 [Google Scholar](#)
- Harterink M, Edwards SL, de Haan B, Yau KW, van den Heuvel S, Kapitein LC, et al (2018) **Local microtubule organization promotes cargo transport in C. elegans dendrites** *J Cell Sci* **131** [Google Scholar](#)
- He L, Kooistra R, Das R, Oudejans E, van Leen E, Ziegler J, et al (2020) **Cortical anchoring of the microtubule cytoskeleton is essential for neuron polarity** *eLife* **9**:e55111 <https://doi.org/10.7554/eLife.55111> | [Google Scholar](#)
- He L, van Beem L, Snel B, Hoogenraad CC, Harterink M (2022) **PTRN-1 (CAMSAP) and NOCA-2 (NINEIN) are required for microtubule polarity in Caenorhabditis elegans dendrites** *PLoS*

Biol **20**:e3001855 [Google Scholar](#)

Herzmann S, Götzelmann I, Reekers L-F, Rumpf S (2018) **Spatial regulation of microtubule disruption during dendrite pruning in *Drosophila*** *Development* **145**:dev156950 [Google Scholar](#)

Herzmann S, Krumkamp R, Rode S, Kintrup C, Rumpf S (2017) **PAR-1 promotes microtubule breakdown during dendrite pruning in *Drosophila*** *EMBO J* **36**:1981–1991 [Google Scholar](#)

Hu C, Feng P, Chen M, Tang Y, Soba P (2022) **Spatiotemporal changes in microtubule dynamics during dendritic morphogenesis** *Fly (Austin)* **16**:13–23 [Google Scholar](#)

Kirilly D, Gu Y, Huang Y, Wu Z, Bashirullah A, Low BC, et al (2009) **A genetic pathway composed of Sox14 and Mical governs severing of dendrites during pruning** *Nat Neurosci* **12**:1497–1505 [Google Scholar](#)

Koch I, Schwarz H, Beuchle D, Goellner B, Langegger M, Aberle H (2008) ***Drosophila* ankyrin 2 is required for synaptic stability** *Neuron* **58**:210–222 [Google Scholar](#)

Kolodziej PA, Jan LY, Jan YN (1995) **Mutations that affect the length, fasciculation, or ventral orientation of specific sensory axons in the *Drosophila* embryo** *Neuron* **15**:273–286 [Google Scholar](#)

Krämer R, Rode S, Rumpf S (2019) **Rab11 is required for neurite pruning and developmental membrane protein degradation in *Drosophila* sensory neurons** *Dev Biol* **451**:68–78 [Google Scholar](#)

Krämer R, Wolterhoff N, Galic M, Rumpf S (2023) **Developmental pruning of sensory neurites by mechanical tearing in *Drosophila*** *J Cell Biol* :222 [Google Scholar](#)

Lee S, Kolodziej PA (2002) **Short Stop provides an essential link between F-actin and microtubules during axon extension** *Development* **129**:1195–1204 [Google Scholar](#)

Liang X, Agulto R, Eichel K, Taylor CA, Paat VA, Deng H, et al (2024) **CRMP/UNC-33 maintains neuronal microtubule arrays by promoting individual microtubule rescue** *bioRxiv* <https://doi.org/10.1101/2024.05.31.596870> | [Google Scholar](#)

Liang X, Kokes M, Fetter RD, Sallee MD, Moore AW, Feldman JL, et al (2020) **Growth cone-localized microtubule organizing center establishes microtubule orientation in dendrites** *eLife* :9 [Google Scholar](#)

Lin T, Kao H-H, Chou C-H, Chou C-Y, Liao Y-C, Lee H-H (2020) **Rab11 activation by Ik2 kinase is required for dendrite pruning in *Drosophila* sensory neurons** *PLoS Genet* **16**:e1008626 [Google Scholar](#)

Marzano M, Herzmann S, Elsbroek L, Sanal N, Tarbashevich K, Raz E, et al (2021) **AMPK adapts metabolism to developmental energy requirement during dendrite pruning in *Drosophila*** *Cell Rep* **37**:110024 [Google Scholar](#)

Matsubara D, Horiuchi S-Y, Shimono K, Usui T, Uemura T (2011) **The seven-pass transmembrane cadherin Flamingo controls dendritic self-avoidance via its binding to a**

LIM domain protein, Espinas, in Drosophila sensory neurons *Genes Dev* **25**:1982–1996 [Google Scholar](#)

Mattie FJ, Stackpole MM, Stone MC, Clippard JR, Rudnick DA, Qiu Y, et al (2010) **Directed microtubule growth, +TIPs, and kinesin-2 are required for uniform microtubule polarity in dendrites.** *Curr Biol* **20**:2169–2177 [Google Scholar](#)

Mukherjee A, Brooks PS, Bernard F, Guichet A, Conduit PT (2020) **Microtubules originate asymmetrically at the somatic golgi and are guided via Kinesin2 to maintain polarity within neurons** *eLife* :9 [Google Scholar](#)

Nashchekin D, Fernandes AR, St Johnston D (2016) **Patronin/Shot Cortical Foci Assemble the Noncentrosomal Microtubule Array that Specifies the Drosophila Anterior- Posterior Axis** *Dev Cell* **38**:61–72 [Google Scholar](#)

Nashchekin D, Squires I, Prokop A, St Johnston D (2024) **The Shot CH1 domain recognises a distinct form of F-actin during Drosophila oocyte determination** *Development* :151 [Google Scholar](#)

Nguyen MM, Stone MC, Rolls MM (2011) **Microtubules are organized independently of the centrosome in Drosophila neurons** *Neural Dev* **6**:38 [Google Scholar](#)

Nguyen MM, McCracken CJ, Milner ES, Goetschius DJ, Weiner AT, Long MK, et al (2014) **γ-tubulin controls neuronal microtubule polarity independently of Golgi outposts** *Mol Biol Cell* **25**:2039–2050 [Google Scholar](#)

Ning W, Yu Y, Xu H, Liu X, Wang D, Wang J, et al (2016) **The CAMSAP3-ACF7 Complex Couples Noncentrosomal Microtubules with Actin Filaments to Coordinate Their Dynamics** *Dev Cell* **39**:61–74 [Google Scholar](#)

Noordstra I, Liu Q, Nijenhuis W, Hua S, Jiang K, Baars M, et al (2016) **Control of apico-basal epithelial polarity by the microtubule minus-end-binding protein CAMSAP3 and spectraplakins ACF7** *J Cell Sci* **129**:4278–4288 [Google Scholar](#)

Okenve-Ramos P, Gosling R, Chojnowska-Monga M, Gupta K, Shields S, Alhadyan H, et al (2024) **Neuronal ageing is promoted by the decay of the microtubule cytoskeleton** *PLoS Biol* **22**:e3002504 [Google Scholar](#)

Ori-McKenney KM, Jan LY, Jan Y-N (2012) **Golgi outposts shape dendrite morphology by functioning as sites of acentrosomal microtubule nucleation in neurons** *Neuron* **76**:921–930 [Google Scholar](#)

Rode S, Ohm H, Anhäuser L, Wagner M, Rosing M, Deng X, et al (2018) **Differential Requirement for Translation Initiation Factor Pathways during Ecdysone-Dependent Neuronal Remodeling in Drosophila** *Cell Rep* **24**:2287–2299 [Google Scholar](#)

Rui M, Ng KS, Tang Q, Bu S, Yu F (2020) **Protein phosphatase PP2A regulates microtubule orientation and dendrite pruning in Drosophila** *EMBO Rep* **21**:e48843 [Google Scholar](#)

Rumpf S, Rode S, Herzmann S, Krumkamp R (2017) **Mechanisms of Neurite Pruning** *e-Neuroforum* **23**:19–25 [Google Scholar](#)

Rumpf S, Wolterhoff N, Herzmann S (2019) **Functions of Microtubule Disassembly during Neurite Pruning** *Trends Cell Biol* **29**:291–297 [Google Scholar](#)

- Sanal N, Keding L, Gigengack U, Michalke E, Rumpf S (2023) **TORC1 regulation of dendrite regrowth after pruning is linked to actin and exocytosis** *PLoS Genet* **19**:e1010526 [Google Scholar](#)
- Schindelin J, Arganda-Carreras I, Frise E, Kaynig V, Longair M, Pietzsch T, et al (2012) **Fiji: an open-source platform for biological-image analysis.** *Nat Methods* **9**:676–682 [Google Scholar](#)
- Slep KC, Rogers SL, Elliott SL, Ohkura H, Kolodziej PA, Vale RD (2005) **Structural determinants for EB1-mediated recruitment of APC and spectraplakins to the microtubule plus end** *J Cell Biol* **168**:587–598 [Google Scholar](#)
- Sun T, Song Y, Dai J, Mao D, Ma M, Ni J-Q, et al (2019) **Spectraplakins Shot Maintains Perinuclear Microtubule Organization in Drosophila Polyploid Cells** *Dev Cell* **49**:731–747 [Google Scholar](#)
- Tang Q, Rui M, Bu S, Wang Y, Chew LY, Yu F (2020) **A microtubule polymerase is required for microtubule orientation and dendrite pruning in Drosophila** *EMBO J* **39**:e103549 [Google Scholar](#)
- Terman JR, Mao T, Pasterkamp RJ, Yu H-H, Kolodkin AL (2002) **MICALs, a family of conserved flavoprotein oxidoreductases, function in plexin-mediated axonal repulsion** *Cell* **109**:887–900 [Google Scholar](#)
- Villars A, Matamoro-Vidal A, Levillayer F, Levayer R (2022) **Microtubule disassembly by caspases is an important rate-limiting step of cell extrusion** *Nat Commun* **13**:3632 [Google Scholar](#)
- Voelzmann A, Liew Y-T, Qu Y, Hahn I, Melero C, Sánchez-Soriano N, et al (2017) **Drosophila Short stop as a paradigm for the role and regulation of spectraplakins** *Semin Cell Dev Biol* **69**:40–57 [Google Scholar](#)
- Voelzmann A, Nuhu-Soso L, Roof A, Patel S, Bennett H, Adamson A, et al (2024) **Mis- regulation of GSK-3 β causes axonal microtubule curling through Shot and Tau** *bioRxiv* :2024.09.08.611864 [Google Scholar](#)
- Wang Y, Rui M, Tang Q, Bu S, Yu F (2019) **Patronin governs minus-end-out orientation of dendritic microtubules to promote dendrite pruning in Drosophila** *eLife* **8** [Google Scholar](#)
- Weiner AT, Seebold DY, Torres-Gutierrez P, Folker C, Swope RD, Kothe GO, et al (2020) **Endosomal Wnt signaling proteins control microtubule nucleation in dendrites** *PLoS Biol* **18**:e3000647 [Google Scholar](#)
- Wolterhoff N, Gigengack U, Rumpf S (2020) **PP2A phosphatase is required for dendrite pruning via actin regulation in Drosophila** *EMBO Rep* **21**:e48870 [Google Scholar](#)
- Yalgin C, Ebrahimi S, Delandre C, Yoong LF, Akimoto S, Tran H, et al (2015) **Centrosomin represses dendrite branching by orienting microtubule nucleation** *Nat Neurosci* **18**:1437–1445 [Google Scholar](#)
- Zheng Y, Wildonger J, Ye B, Zhang Y, Kita A, Younger SH, et al (2008) **Dynein is required for polarized dendritic transport and uniform microtubule orientation in axons** *Nat Cell Biol* **10**:1172–1180 [Google Scholar](#)

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Reviewer #1 (Public review):

Summary:

The Neuronal microtubule cytoskeleton is essential long long-range transport in axons and dendrites. The axon-specific plus-end out microtubule organization vs the dendritic-specific plus-end in organization allows for selective transport into each neurite, setting up neuronal polarity. In addition, the dendritic microtubule organization is thought to be important for

dendritic pruning in *Drosophila* during metamorphosis. However, the precise mechanisms that organize microtubules in neurons are still incompletely understood.

In the current manuscript, the authors describe the spectraplakin protein Shot as important in developmental dendritic pruning. They find that Shot has dendritic microtubule polarity defects, which, based on their rescues and previous work, is likely the reason for the pruning defect.

Since Shot is a known actin-microtubule crosslinker, they also investigate the putative role of actin and find that actin is also important for dendritic pruning. Finally, they find that several factors that have been shown to function as a dendritic MTOC in *C. elegans* also show a defect in *Drosophila* upon depletion.

Strengths:

Overall, this work was technically well-performed, using advanced genetics and imaging. The author reports some interesting findings identifying new players for dendritic microtubule organization and pruning.

Weaknesses:

The evidence for Shot interacting with actin for its functioning is contradictory. The Shot lacking the actin interaction domain did not rescue the mutant; however, it also has a strong toxic effect upon overexpression in wildtype (Figure S3), so a potential rescue may be masked. Moreover, the C-terminus-only construct, which carries the GAS2-like domain, was sufficient to rescue the pruning. This actually suggests that MT bundling/stabilization is the main function of Shot (and no actin binding is needed). On the other hand, actin depolymerization leads to some microtubule defects and subtle changes in shot localization in young neurons (not old ones). More importantly, it did not enhance the microtubule or pruning defects of the Shot domain, suggesting these act in the same pathway. Interesting to note is that Mical expression led to microtubule defects but not to pruning defects. This argues that MT organization effects alone are not enough to cause pruning defects. This may be good to discuss. For the actin depolymerization, the authors used overexpression of the actin-oxidizing Mical protein. However, Mical may have another target. It would be good to validate key findings with better characterized actin targeting tools.

In analogy to *C. elegans*, where RAB-11 functions as a ncMTOC to set up microtubules in dendrites, the authors investigated the role of these in *Drosophila*. Interestingly, they find that rab-11 also colocalizes to gamma tubulin and its depletion leads to some microtubule defects. Furthermore, they find a genetic interaction between these components and Shot; however, this does not prove that these components act together (if at all, it would be the opposite). This should be made more clear. What would be needed to connect these is to address RAB-11 localization + gamma-tubulin upon shot depletion.

All components studied in this manuscript lead to a partial reversal of microtubules in the dendrite. However, it is not clear from how the data is represented if the microtubule defect is subtle in all animals or whether it is partially penetrant stronger effect (a few animals/neurons have a strong phenotype). This is relevant as this may suggest that other mechanisms are also required for this organization, and it would make it markedly different from *C. elegans*. This should be discussed and potentially represented differently.

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Reviewer #2 (Public review):

Summary:

In their manuscript, the authors reveal that the spectraplakins Shot, which can bind both microtubules and actin, is essential for the proper pruning of dendrites in a developing *Drosophila* model. A molecular basis for the coordination of these two cytoskeletons during neuronal development has been elusive, and the authors' data point to the role of Shot in regulating microtubule polarity and growth through one of its actin-binding domains. The authors also propose an intriguing new activity for a spectraplakins: functioning as part of a microtubule-organizing center (MTOC).

Strengths:

(1) A strength of the manuscript is the authors' data supporting the idea that Shot regulates dendrite pruning via its actin-binding CH1 domain and that this domain is also implicated in Shot's ability to regulate microtubule polarity and growth (although see comments below); these data are consistent with the authors' model that Shot acts through both the actin and microtubule cytoskeletons to regulate neuronal development.

(2) Another strength of the manuscript is the data in support of Rab11 functioning as an MTOC in young larvae but not older larvae; this is an important finding that may resolve some debates in the literature. The finding that Rab11 and Msps coimmunoprecipitate is nice evidence in support of the idea that Rab11(+) endosomes serve as MTOCs.

Weaknesses:

(1) A significant, major concern is that most of the authors' main conclusions are not (well) supported, in particular, the model that Shot functions as part of an MTOC. The story has many interesting components, but lacks the experimental depth to support the authors' claims.

(2) One of the authors' central claims is that Shot functions as part of a non-centrosomal MTOC, presumably a MTOC anchored on Rab11(+) endosomes. For example, in the Introduction, last paragraph, the authors summarize their model: "Shot localizes to dendrite tips in an actin-dependent manner where it recruits factors cooperating with an early-acting, Rab11-dependent MTOC." This statement is not supported. The authors do not show any data that Shot localizes with Rab11 or that Rab11 localization or its MTOC activity is affected by the loss of Shot (or otherwise manipulating Shot). A genetic interaction between Shot and Rab11 is not sufficient to support this claim, which relies on the proteins functioning together at a certain place and time. On a related note, the claim that Shot localization to dendrite tips is actin-dependent is not well supported: the authors show that the CH1 domain is needed to enrich Shot at dendrite tips, but they do not directly manipulate actin (it would be helpful if the authors showed the overexpression of Mical disrupted actin, as they predict).

(3) The authors show an image that Shot colocalizes with the EB1-mScarlet3 comet initiation sites and use this representative image to generate a model that Shot functions as part of an MTOC. However, this conclusion needs additional support: the authors should quantify the frequency of EB1 comets that originate from Shot-GFP aggregates, report the orientation of EB1 comets that originate from Shot-GFP aggregates (e.g., do the Shot-GFP aggregates correlate with anterogradely or retrogradely moving EB1 comets), and characterize the developmental timing of these events. The genetic interaction tests revealing ability of shot dsRNA to enhance the loss of microtubule-interacting proteins (Msps, Patronin, EB1) and Rab11 are consistent with the idea that Shot regulates microtubules, but it does not provide any spatial information on where Shot is interacting with these proteins, which is critical to the model that Shot is acting as part of a dendritic MTOC.

(4) It is unclear whether the authors are proposing that dendrite pruning defects are due to an early function of Shot in regulating microtubule polarity in young neurons (during 1st instar larval stages) or whether Shot is acting in another way to affect dendrite pruning. It

would be helpful for the authors to present and discuss a specific model regarding Shot's regulation of dendrite pruning in the Discussion.

(5) The authors argue that a change in microtubule polarity contributes to dendrite pruning defects. For example, in the Introduction, last paragraph, the authors state: "Loss of Shot causes pruning defects caused by mixed orientation of dendritic microtubules." The authors show a correlative relationship, not a causal one. In Figure 4, C and E, the authors show that overexpression of Mical disrupts microtubule polarity but not dendrite pruning, raising the question of whether disrupting microtubule polarity is sufficient to cause dendrite pruning defects. The lack of an association between a disruption in microtubule polarity and dendrite pruning in neurons overexpressing Mical is an important finding.

(6) The authors show that a truncated Shot construct with the microtubule-binding domain, but no actin-binding domain (Shot-C-term), can rescue dendrite pruning defects and Khc-lacZ localization, whereas the longer Shot construct that lacks just one actin-binding domain ("delta-CH1") cannot. Have the authors confirmed that both proteins are expressed at equivalent levels? Based on these results and their finding that over-expression of Shot-delta-CH1 disrupts dendrite pruning, it seems possible that Shot-delta-CH1 may function as a dominant-negative rather than a loss-of-function. Regardless, the authors should develop a model that takes into account their findings that Shot, without any actin-binding domains and only a microtubule-binding domain, shows robust rescue.

(7) The authors state that: "The fact that Shot variants lacking the CH1 domain cannot rescue the pruning defects of shot[3] mutants suggested that dendrite tip localization of Shot was important for its function." (pages 10-11). This statement is not accurate: the Shot C-term construct, which lacks the CH1 domain (as well as other domains), is able to rescue dendrite pruning defects.

(8) The authors state that: "In further support of non-functionality, overexpression of Shot[deltaCH1] caused strong pruning defects (Fig. S3)." (page 8). Presumably, these results indicate that Shot-delta-CH1 is functioning as a dominant-negative since a loss-of-function protein would have no effect. The authors should revise how they interpret these results. This comment is related to another comment about the ability of Shot constructs to rescue the shot[3] mutant.

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